Lufthansa

Project Goals

The primary goals of this project were to:

- Analyze flight delays and cancellations at Lufthansa Airlines
- Identify key operational bottlenecks
- Distinguish between true delay vs. process delay
- Understand peak vs. off-peak operational pressure
- Build a decision-ready Power BI dashboard for management use

Project Introduction

- Data modeling and transformation
- DAX-based analytical measures
- Power BI dashboard design
- Business-oriented insight generatio

Key Achievements

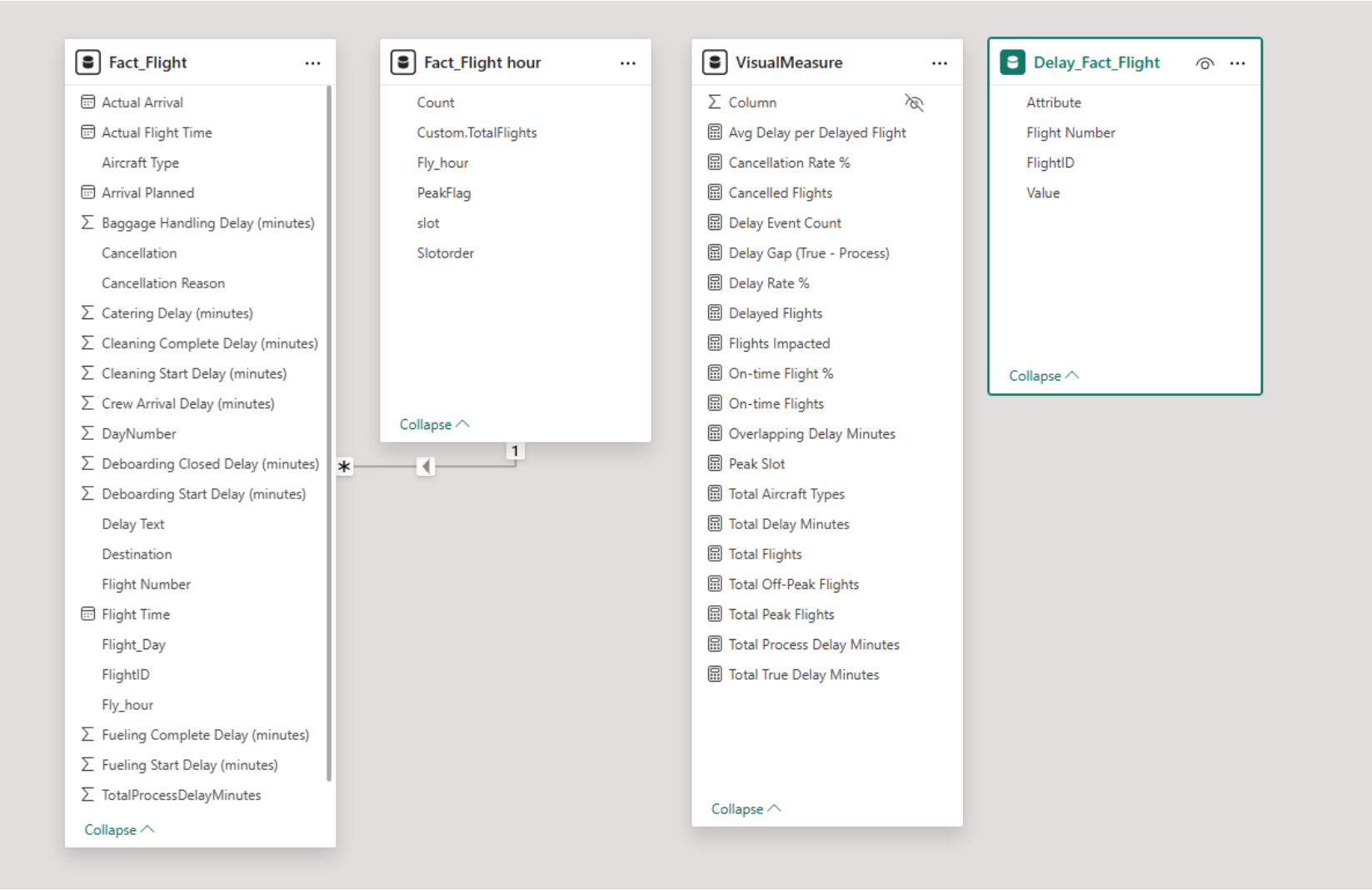
✓ Built a structured fact-based data model
✓ Developed advanced DAX measures for delay and performance analysis
✓ Identified peak operational hours impacting delays
✓ Designed interactive Power BI dashboards
✓ Converted raw data into actionable business insights

KPI Summary (Dashboard Overview

- Total Flights
- Delayed Flights
- Cancelled Flights
- On-Time Flights
- Total True Delay Minutes
- Total Process Delay Minutes

PESTEL Analysis
Lufthansa Airlines

- **Political:** EU aviation rules, ATC regulations, and strikes strongly affect punctuality and cancellations.
- **Economic:** Fuel price volatility and delay compensation costs (EU261) directly impact profitability.
- **Social:** High passenger expectations for on-time performance; delays quickly affect brand trust.
- **Technological:** Advanced analytics (Power BI, AI) improve delay detection, recovery, and scheduling efficiency.
- **Environmental:** Weather disruptions and sustainability pressure increase operational complexity.
- **Legal:** Strict EU passenger-rights laws increase financial and compliance risk for delays and cancellations



Data Modeling Approach

- The analysis is built on a central **Fact_Flight** table containing flight-level operational data.
- Supporting analytical tables were created to enable time-based and peak/off-peak analysis.
- Calculated measures (DAX) were used for KPIs such as delays, cancellations, on-time rate, and true vs process delay.
- The model follows a **fact-driven analytical design**, optimized for performance and business insight generation.

1680 Total Flights	1178 On-time Flights	374 Delayed Flights	128 Cancelled Flights
	70,12 On-time Flight %	22,26 Delay Rate %	7,62 Cancellation Rat...

Operational Volume & Reliability
Total Flights: 1,680
On-time Flights: 1,178 (**70.12% on-time performance**)
Delayed Flights: 374 (**22.26% delay rate**)
Cancelled Flights: 128 (**7.62% cancellation rate**)
"Out of 1,680 Lufthansa flights, just over 70% operated on time. Approximately 22% experienced delays, while cancellations remained below 8%, indicating overall operational stability with clear opportunities to improve punctuality."

35152 Total True Delay Minutes	170749 Total Process Delay Minutes	936 Total Off-Peak Flights	744 Total Peak Flights

Delay Severity: Passenger vs Internal Impact
Total True Delay Minutes (Passenger Impact): 35,152
Total Process Delay Minutes (Internal Operations): 170,749
"A key insight is the significant gap between internal process delays and passenger-experienced delays. While Lufthansa accumulates over 170,000 minutes of internal operational delays, only about 35,000 minutes are ultimately passed on to passengers."
This indicates strong delay-absorption capability through:
Buffer time management
Schedule flexibility
Operational recovery mechanisms

Interpretation of the Statistics

•A 7.62% cancellation rate indicates that cancellations are not negligible, but they are significantly lower than delay occurrences (22.26%).

•The majority of operational disruptions are therefore handled as delays rather than cancellations, suggesting:

- *Effective recovery and re-scheduling strategies*
- *A preference for absorbing disruption operationally instead of cancelling flights outright*

Total Flight : 1680
Total Cancel Flight : 128
Cancel ratio: 7.62%

Comparative Context

•While **22.26% of flights experience delays**, only **7.62% are cancelled**, meaning:

- Many disrupted flights are **successfully recovered**
- Operational buffers and contingency planning prevent escalation from delay to cancellation

Business Impact

•Positive:

- Lower cancellation rates protect customer trust and brand reputation
- Reduced compensation and re-accommodation costs compared to cancellations

•Risk Area:

- Each cancelled flight has a disproportionately high cost impact compared to delayed flights
- Even small reductions in cancellations can yield large financial and customer-satisfaction gains

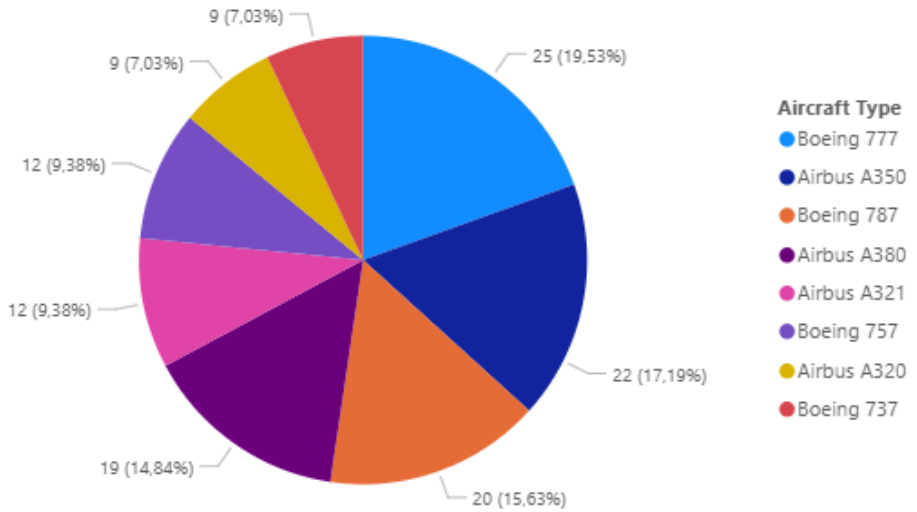
Risk Area

•Each cancelled flight carries a **disproportionately high cost impact** compared to delayed flights:

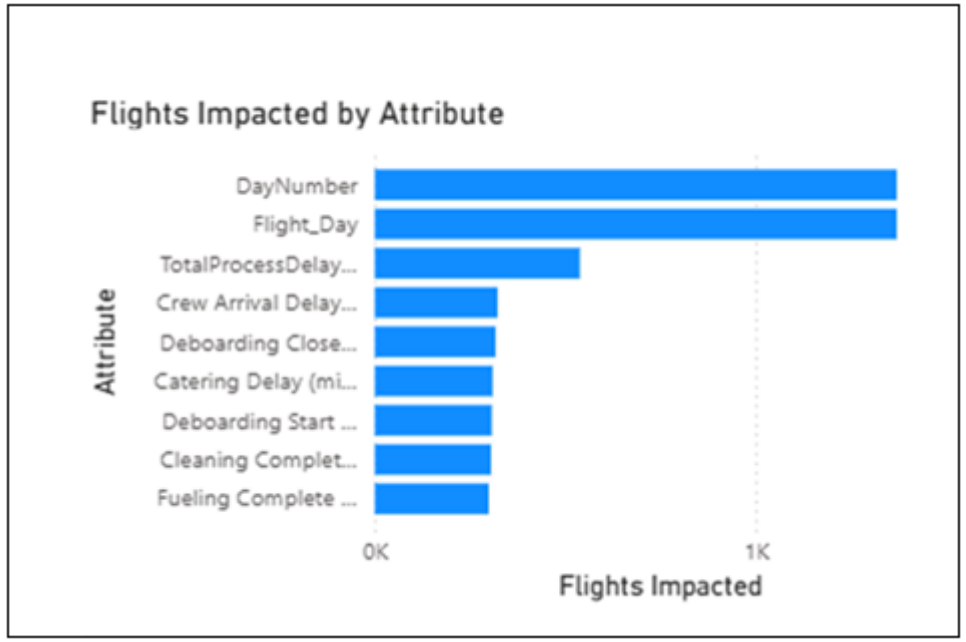
- Passenger dissatisfaction
- Missed connections
- Regulatory and compensation exposure

•Even **small reductions in cancellations** can generate **outsized financial and customer-experience gains**

Cancelled Flights by Aircraft Type

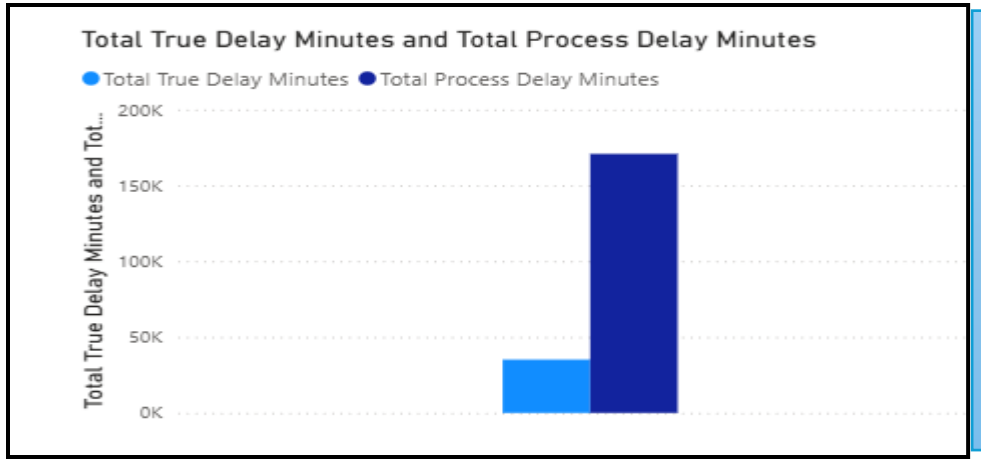


"Although Lufthansa operates in a disruption-prone environment, only 7.62% of flights are cancelled. This shows strong operational control, with most disruptions being managed through delays rather than cancellations. Further reducing cancellations—even marginally—would deliver significant cost savings and customer experience benefits."



Flights Impacted by Attribute:

- This chart highlights the operational and scheduling attributes that have impacted the highest number of Lufthansa flights.
- The strongest impact is driven by day-level patterns and overall process-related delays, while individual operational activities such as crew arrival, catering, deboarding, and cleaning contribute comparatively less.
- This clearly indicates that flight delays are systemic and pattern-based, rather than the result of isolated operational incidents.

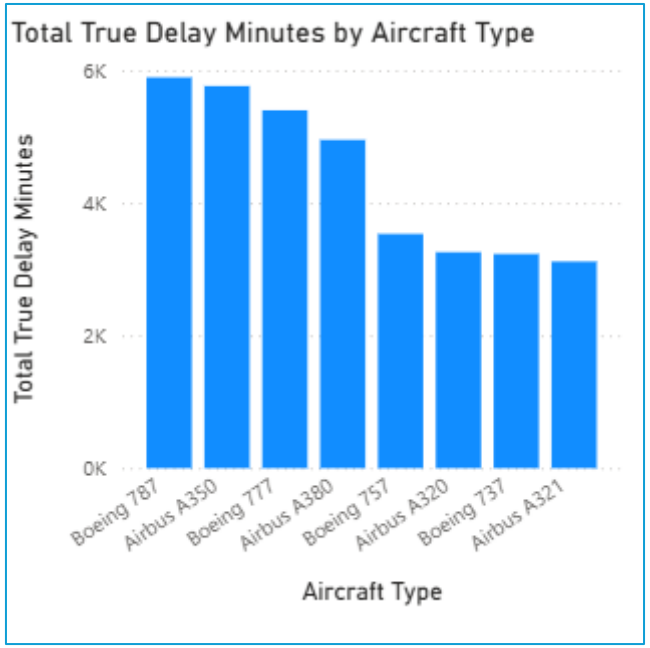


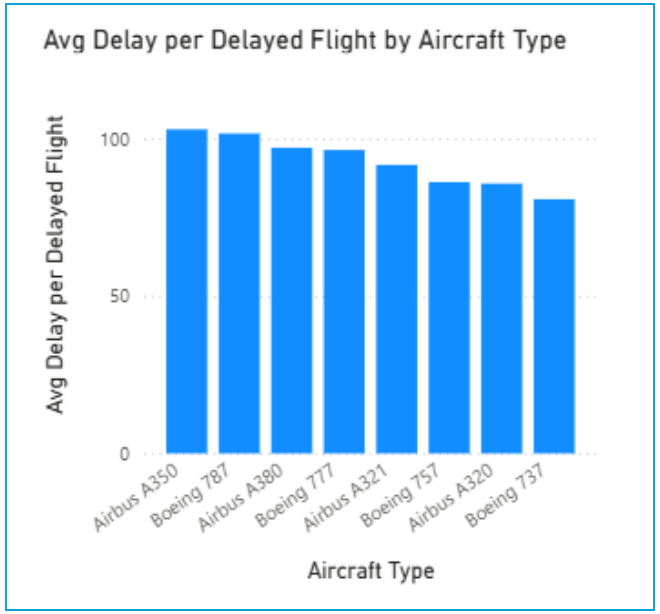
Total True Delay Minutes vs Total Process Delay Minutes

- This chart compares the actual delay experienced by passengers with the total internal process-related delay minutes.
- Although total process delays are significantly higher, only a portion of that delay translates into true passenger delay.
  This demonstrates Lufthansa's ability to absorb operational inefficiencies through buffer time, scheduling flexibility, and recovery mechanisms.

Business Impact

- Operational efficiency opportunity: High process-related delays point to inefficiencies in ground operations and aircraft turnaround activities.
- Cost reduction potential: Optimizing internal processes can reduce operational costs, even when passenger delays are already partially mitigated.
- Service quality protection: Effective delay absorption helps preserve customer satisfaction and maintain on-time performance standards.
- Strategic optimization focus: Process-level improvements deliver higher impact than addressing isolated delay causes.





Executive Explanation:

This slide analyzes delay impact by aircraft type from two complementary perspectives: total passenger delay minutes and average delay per delayed flight.

Wide-body aircraft—specifically the Boeing 787, Airbus A350, and Boeing 777—generate the highest total delay minutes. This indicates that delays affecting long-haul fleets have a broader passenger impact due to longer flight durations and higher passenger volumes.

At the same time, these aircraft also show a higher average delay per delayed flight, confirming that when delays occur on wide-body aircraft, they tend to be more severe rather than short, recoverable disruptions.

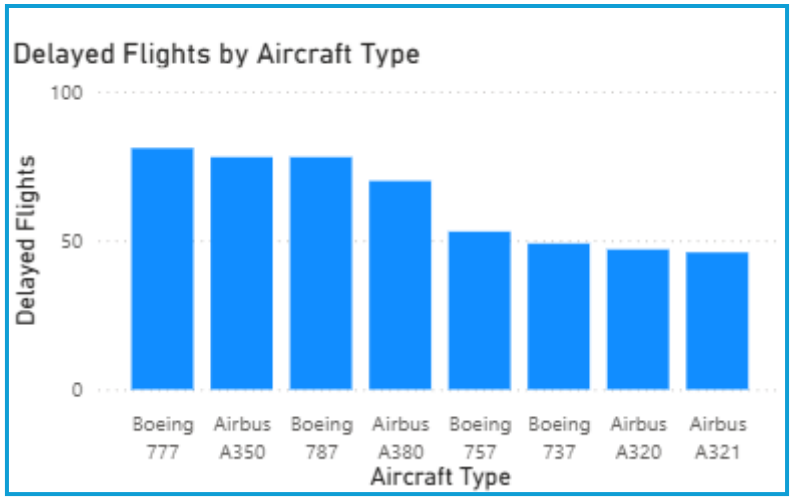
Overall, this demonstrates that operational issues impacting long-haul fleets have a disproportionately large effect on Lufthansa's total delay performance.

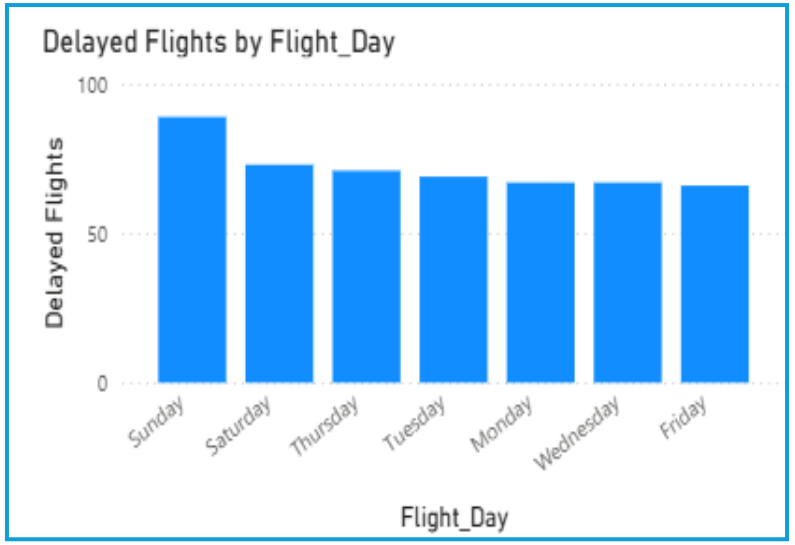
Business Insight (Executive-Focused):

Primary optimization opportunity: Wide-body aircraft turnaround, maintenance coordination, and crew readiness

High leverage area: Even small efficiency improvements on long-haul fleets can lead to significant passenger delay reduction

Strategic implication: Delay-reduction initiatives should prioritize aircraft types with the highest combined impact × severity



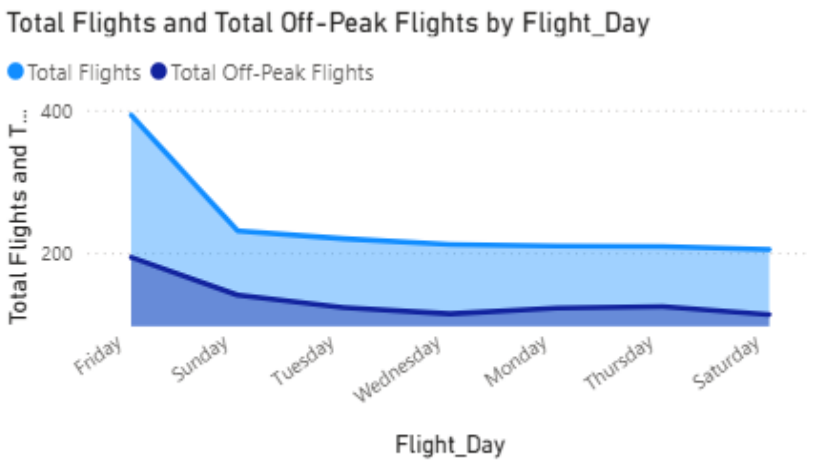


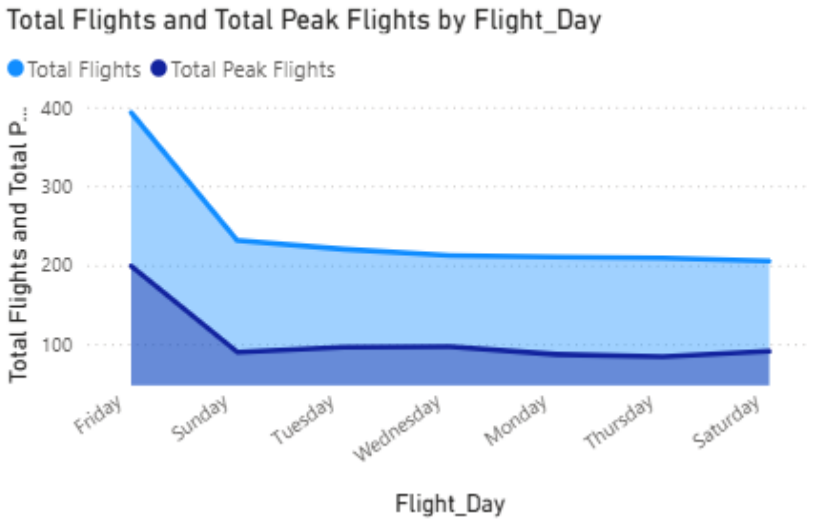
Executive Explanation:

- This slide analyzes how frequently delays occur, focusing on two dimensions: aircraft type and day of the week.
- The upper chart shows that wide-body aircraft—particularly the Boeing 777, Airbus A350, and Boeing 787—experience the highest number of delayed flights. These aircraft are heavily utilized on long-haul routes and typically operate during high-demand time windows, increasing their exposure to operational disruptions.
- The lower chart highlights day-level delay patterns, where weekend and early-week days show a higher concentration of delayed flights. This suggests that traffic density, passenger volume, and network complexity vary significantly by day, influencing delay risk.
- Together, these charts indicate that delay frequency is not random but strongly influenced by fleet composition and scheduling patterns.

Business Insight:

- Key risk factor: Peak-hour and peak-day utilization of wide-body aircraft
- Operational opportunity:
  - Smoother schedule distribution across days
  - Additional buffer time for long-haul aircraft during high-risk periods
- Strategic takeaway: Reducing delay frequency requires capacity-aware scheduling, not only reactive delay management





Total Flights vs Off-Peak Flights by Flight Day (Top Chart)

Explanation:

This chart illustrates the distribution of total flights compared with off-peak flights across different days of the week.

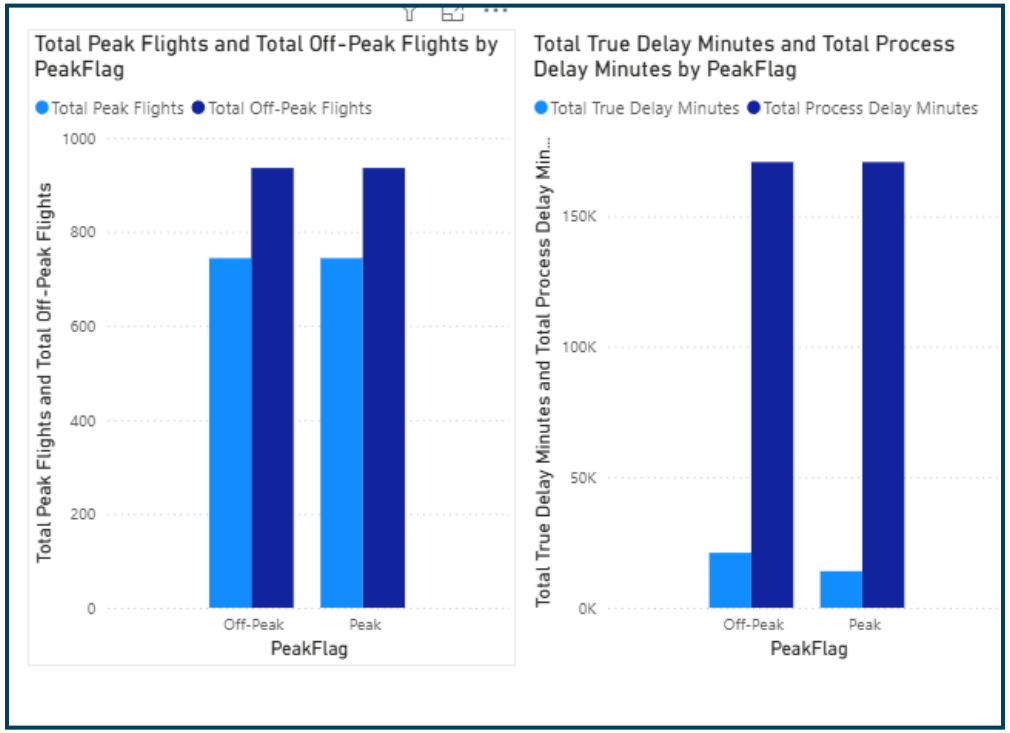
- Friday shows the highest overall flight volume, indicating strong end-of-week travel demand.
- After Friday, total flight volume drops sharply on Sunday and then stabilizes during weekdays.
- Off-peak flights follow a similar pattern but remain consistently lower than total flights, reflecting structured scheduling outside peak hours.

Total Flights vs Peak Flights by Flight Day (Bottom Chart)

Explanation:

This chart compares total flights with peak-hour flights across the week.

- Peak flights are most concentrated on Friday, aligning with high passenger demand.
- After the weekend, peak flight volumes decrease and remain relatively flat during weekdays.
- The gap between total flights and peak flights highlights that only a portion of daily operations occur during peak hours, even on high-traffic days.



Total Peak vs Off-Peak Flights (Left Chart)

This chart compares the volume of flights operated during peak and off-peak hours.

- Off-peak flights are slightly higher in number than peak flights.
- Peak flights represent a smaller share of total operations, but they are strategically important because they occur during high-demand time windows.

True Delay Minutes vs Process Delay Minutes (Right Chart)

This chart compares:

- Total Process Delay Minutes (internal operational delays), and
- Total True Delay Minutes (actual passenger-experienced delays).

Key observations:

- Process delay minutes are extremely high in both peak and off-peak periods.
- However, true passenger delay minutes are significantly lower, especially during peak hours.

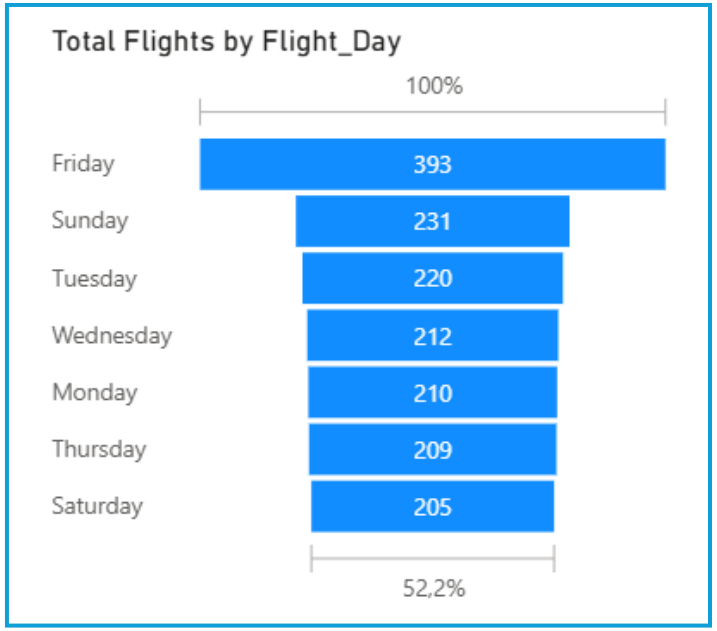
Interpretation:

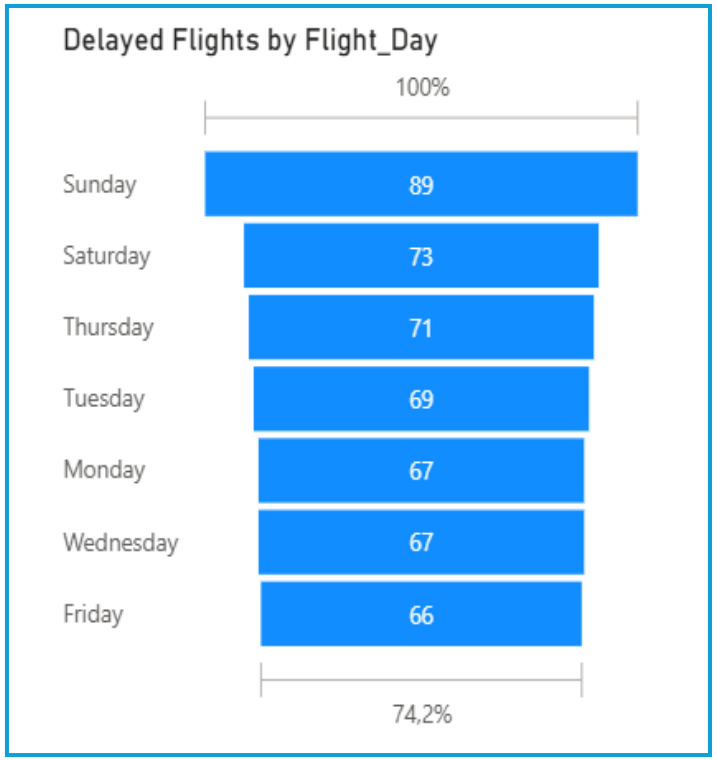
Although internal operational delays increase during peak hours, only a portion of that delay is passed on to passengers.

Business Impact (Use This Section as Highlight)

- Operational strength: Lufthansa demonstrates strong delay-absorption capability during peak hours
- Risk indicator: High internal process delays signal hidden inefficiencies in ground operations and turnaround activities
- Optimization opportunity:

Reducing process-level delays—even slightly—can significantly lower operational cost without impacting customer experience





Total Flights by Flight Day (Left / Top Chart)

Explanation:

This chart shows the total number of Lufthansa flights operated on each day of the week.

Key Observations:

•Friday records the highest number of flights (393), indicating strong end-of-week travel demand.

•After Friday, total flight volume drops sharply on Sunday (231) and then remains relatively stable during weekdays.

•Saturday has the lowest total flight volume (205), reflecting reduced business and commuter travel.

Interpretation:

Lufthansa's weekly schedule is clearly demand-driven, with capacity peaking toward the end of the workweek and stabilizing during regular weekdays.

Delayed Flights by Flight Day (Right / Bottom Chart)

Explanation:

This chart shows the number of delayed flights for each day of the week.

Key Observations:

•Sunday has the highest number of delayed flights (89), despite not having the highest total flight volume.

•Friday, although the busiest day, has the lowest delayed flights (66).

•Delay counts remain relatively high on Saturday and Thursday, indicating operational pressure during weekends.

Interpretation:

Flight delays are not directly proportional to flight volume.

Instead, they appear to be influenced by operational recovery challenges, weekend scheduling constraints, and aircraft/crew rotation patterns.

Story (after the visual):

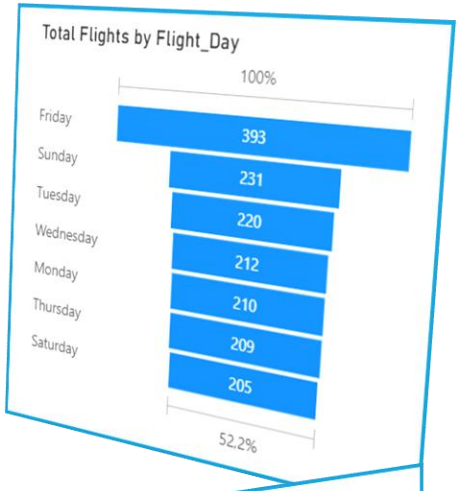
"Although Friday has the highest number of flights, it records the lowest delays. Sunday, despite lower traffic, experiences the highest delays—indicating recovery challenges rather than capacity issues."

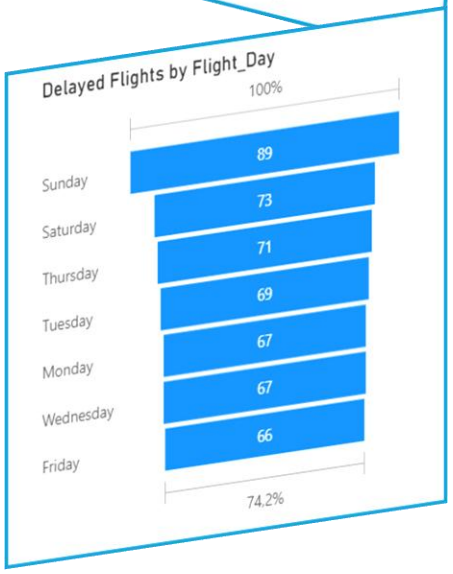
Combined Insight (Very Important for Presentation)
High traffic does not automatically mean high delays.

•Lufthansa performs **operationally strongest on Fridays**, managing the highest traffic with comparatively fewer delays.

•**Sundays represent a risk day**, where fewer flights still result in more delays—suggesting:

- tighter turnaround buffers,
- accumulated delays from the week,
- limited recovery windows before the new week begins.





Business Impact & Recommendations
Operational Insight:

•Fridays demonstrate best-practice scheduling and buffer management.
•Sundays expose systemic recovery and turnaround inefficiencies.

Actionable Recommendations:
•Reallocate buffer time, crew availability, and ground resources toward Sunday operations.
•Apply Friday-style scheduling discipline to weekend operations.
•Focus improvement efforts on day-level planning, not just peak-hour optimization.

Strategic Value:
•Small operational improvements on Sundays can significantly reduce weekly delay totals.
•Better recovery planning improves on-time performance without increasing capacity.

Short Storytelling
Lufthansa Delay & Cancellation Analysis

This project analyzes Lufthansa's flight delays and cancellations to understand where operational risks truly lie.
The key insight is that high traffic does not automatically mean high delays. Lufthansa performs strongest on Fridays, managing the highest flight volume with the lowest delay impact. In contrast, Sundays emerge as a risk day, with fewer flights but more delays—highlighting recovery and turnaround challenges rather than capacity issues.
Despite significant internal process delays, Lufthansa successfully absorbs most disruptions, keeping cancellations low at 7.62%, demonstrating strong operational resilience with clear opportunities for targeted optimization.

"This project is an independent analytical case study based on publicly available / simulated data and is not an official Lufthansa internal report."